Jidu Rothschild

1. $SE(\bar{x}) = \frac{10}{\sqrt{100}} = 1$

$z = \frac{52-50}{1} = 2$

$\alpha = 0.1$ $\alpha = 0.05$ $\alpha = 0.01$

$2 > 1.645$ $2 > 1.96$ $2 < 2.576$

Sig - reject Sig - reject Not sig → fail to reject

$52 \pm 1.96 \times 1 = (50.04, 53.96)$ → 95% CI

2. a) $1.645 \times 1.3 = 2.1385$ (0.9615, 5.2385) : reject

b) $1.96 \times 1.3 = 2.548$ (0.752, 5.648) : reject

c) $2.576 \times 1.3 = 3.3488$ (-0.2488, 6.4488) : fail to reject

3. $SE = \frac{1}{2}$ $1.96 \times 0.5 = 0.98$

11 ± 0.98: (10.52, 12.48) 95% CI. Yes fail to reject H_0.

4. $SE = \sqrt{\frac{4}{100} + \frac{42}{100}} = \sqrt{1.13} \approx 1.06$ $1.06 \times 1.95 \approx 2.08$

$25 - 22 = 3 \pm 2.08 = (0.92, 5.08)$ reject H_0 at 95% confidence

$z = 2.08$

5. $SE = \sqrt{\frac{36}{81} + \frac{25}{64}} \approx 0.91$

6. $\sqrt{\frac{0.5}{40} + \frac{1.21}{64}} \approx 0.23 \times 1.96 = 0.47$ $z = 0.47$

$0.4 \pm 0.47 = (-0.07, 0.87)$ fail to reject H_0. Not statistically significant.